CLIMATE CHANGE PUBLIC OPINION ANALYSIS

A Twitter Data Analytics Report

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Tools Used: PostgreSQL, Power BI

Dataset: The Climate Change Twitter Dataset (Mendeley Data, 2022)

1. INTRODUCTION

This report presents a comprehensive end-to-end data analysis of global public opinion on climate change using over 15 million tweets spanning from 2006 to 2019. The key objectives were to understand how public sentiment and stance have shifted over 13 years, and to identify which topics and regions are driving the most divisive or aggressive discourse on climate change. The analysis was conducted using PostgreSQL for data preparation and transformation, and Power BI for interactive visualisation.

2. DATA UNDERSTANDING

2.1 Dataset Overview

The dataset was sourced from Mendeley Data and contains 15,789,411 tweets collected between June 2006 and October 2019. Each tweet is tagged with metadata including sentiment scores, climate change stance, topic category, geolocation coordinates, author gender, temperature deviation, and aggressiveness labels.

2.2 Data Dictionary

Column	Data Type	Description
id	BIGINT	Unique tweet identifier
created_at	TIMESTAMP	Date and time tweet was posted
lat	NUMERIC	Latitude coordinate of tweet origin
lng	NUMERIC	Longitude coordinate of tweet origin

topic	TEXT	Climate topic category of the tweet
sentiment	NUMERIC	Sentiment score ranging from -1 (negative) to 1 (positive)
stance	TEXT	Author's climate stance: believer, denier, or neutral
gender	TEXT	Inferred author gender: male, female, or unknown
temperature_avg	NUMERIC	Average temperature deviation at the tweet location
aggressiveness	TEXT	Whether a tweet is aggressive or not aggressive

3. DATA PREPARATION & CLEANING

3.1 Approach

All data preparation was performed in PostgreSQL. The raw CSV file was loaded into a table with all columns initially stored as TEXT to handle formatting inconsistencies across 15 million rows safely. Columns were then converted to their appropriate data types using ALTER TABLE statements.

3.2 Data Quality Issues Found

Issue	Finding	Action Taken
Missing coordinates	10,481,873 rows (66%) had null lat/lng	Retained as NULL, flagged with has_location column
Missing temperature	10,481,873 rows (66%) had null temperature_avg	Retained as NULL , flagged with has_temperature column
Undefined gender	Present in the dataset	Standardised to "unknown."
Duplicate tweet IDs	0 duplicates found	No action needed
Sentiment out of range	0 values outside -1 to 1	No action needed
Mixed date formats	Two formats are present in created_at	Handled with CASE statement during type conversion
Inconsistent text casing	Stance, gender, and aggressiveness had mixed cases	Standardised with TRIM(LOWER())

3.3 Cleaned Dataset

After cleaning, the tweets_clean table was created containing all 15,789,411 rows with properly typed columns, standardised categorical values, and two additional boolean flag columns for location and temperature availability.

4. DESCRIPTIVE ANALYTICS

4.1 Summary Statistics

Metric	Value
Total Tweets	15,789,411
Date Range	June 2006 — October 2019
Average Sentiment Score	0.0025
Believers	11M (71.52%)
Neutrals	3M (20.94%)
Deniers	1M (7.55%)
Aggressive Tweets	5M (28.67%)
Non-Aggressive Tweets	11M (71.33%)

4.2 Key Observations

Sentiment: The overall average sentiment of 0.0025 indicates that public discourse on climate change is essentially neutral on aggregate, masking significant variation across topics, stances, and time periods.

Stance Distribution: Believers dominate the dataset at 71.52%, suggesting that the majority of Twitter users discussing climate change accept its reality. Deniers represent only 7.55% of the conversation, though their impact on aggressiveness is disproportionate.

Aggressiveness: Nearly 1 in 3 tweets (28.67%) were classified as aggressive, highlighting the highly charged nature of climate change discourse on social media.

Gender: Male authors account for 65.28% of tweets, female authors 31%, and unknown gender 3.72%, reflecting the broader gender gap in political and scientific discourse on Twitter.

5. DIAGNOSTIC ANALYTICS

5.1 Sentiment Trend Over Time

Sentiment started relatively positive in 2006–2007 at around 0.1, then dropped sharply between 2008 and 2010 to approximately -0.08. Sentiment gradually recovered between 2013 and 2016, reaching a local peak around 0.05, before declining again toward 2019. This pattern suggests that major global events and political developments have a measurable impact on public sentiment toward climate change.

5.2 Topic Analysis

Politics is the most aggressive topic, followed closely by Donald Trump versus Science and Ideological Positions on Global Warming. These three topics consistently generate the highest aggressiveness percentages, indicating that politically charged climate topics provoke the most hostile responses. In contrast, topics like Significance of Pollution and Seriousness of Gas Emissions generate more measured discourse.

Ideological Positions on Global Warming shows the most negative average sentiment, suggesting deep frustration or conflict around ideological debates. Global stance and Importance of Human Intervention show more positive sentiment, indicating that constructive framing of climate topics generates more positive responses.

5.3 Stance vs Topic

Global stance is the dominant topic across all three stance categories, with believers contributing the largest share. The Donald Trump versus Science topic shows a notably higher proportion of deniers compared to other topics, confirming the political polarisation of climate discourse during the Trump era.

5.4 Gender Analysis

Male authors dominate tweet volume across all stance categories. However, female authors show a higher proportion of believers relative to their total volume compared to male authors, suggesting gender differences in climate change acceptance. Unknown gender accounts for a very small share across all stances.

5.5 Temperature vs Sentiment

The temperature versus sentiment chart reveals that extreme temperature deviations, both very low and very high, correlate with more volatile sentiment scores. At moderate temperature deviations near zero, sentiment stabilises around neutral. This suggests that direct experience of extreme weather may influence emotional responses in climate change discourse.

6. RECOMMENDATIONS

1. Target Politically Charged Topics for Intervention: Politics, Donald Trump versus Science, and Ideological Positions on Global Warming are the most aggressive topics. Communication strategies should focus on de-escalating discourse in these areas.

2. Leverage Believer Majority: With 71.52% of the conversation coming from believers, climate organisations have a strong base to amplify. Campaigns should mobilise this majority more effectively rather than focusing resources on converting deniers.

3. Address the Gender Gap: The significant underrepresentation of female voices (31% vs 65% male) suggests an opportunity to create more inclusive platforms and targeted outreach to increase female engagement in climate discourse.

4. Monitor Sentiment Around Key Events: The clear correlation between global events and sentiment dips indicates that real-time sentiment monitoring around major political events, summits, and climate reports would provide valuable early warning signals for communication teams.

6. Invest in Geolocation Data: With 66% of tweets missing location data, geographic analysis is currently limited. Future data collection strategies should prioritise location enrichment to enable more precise regional targeting of climate communication campaigns.

7. CONCLUSIONS

This analysis of 15.7 million climate change tweets reveals a public discourse that is dominated by believers but deeply divided along political lines. Sentiment has fluctuated significantly over 13 years in response to major global events, and nearly one-third of all climate tweets contain aggressive language. The most divisive topics are politically charged, particularly those involving Donald Trump and ideological debates, while constructive topics generate more positive engagement.

8. REFERENCES

Qian, Y., & Goldsmith, D. (2022). *The Climate Change Twitter Dataset*. Mendeley Data, V2. https://data.mendeley.com/datasets/mw8yd7z9wc/2